

UNIT V – ADVANCED MATERIALS

5.1 INTRODUCTION

Materials form an integral part of our life. Advancement in our day to day life leads to the discovery of many new Engineering materials such as metallic glasses, Nano-phase materials, ceramic, shape-memory alloys etc.

All these materials are composite materials which are prepared by grouping one type of materials with the other; say for example, metals mixed with polymer, ceramic mixed with metals etc. Also it was found that the resultant materials have some peculiar properties, which varies with respect to size, shape and the nature of mixture. Because of their peculiar properties these materials are used in high-tech applications such as magnetic alloys in CD's /DVD's, Titanium alloys in supersonic planes, space shuttles etc. In this chapter some of the new engineering materials along with their properties and the wide range of application it has in this Modern Engineering world were discussed.

5. 2 METALLIC GLASSES

Metallic glasses are new type of materials which possess both the properties of metals and glasses are listed below in the table.

Properties of metals

+

= **Properties of metallic glasses**

Properties of glass

Sl.No	Properties of metals	Properties of glasses	Properties of metallic glasses
1.	Crystalline	Amorphous	Non-crystalline and amorphous alloys.
2.	Ductile	Brittle	Brittle and ductile
3.	Not-transparent(opaque) Malleable	Transparent	Malleable and opaque

Definition:

Metallic glasses are the amorphous solids which have high strength, good magnetic properties and better corrosion resistance and will possess both the properties of metals and glasses.

Example: Alloys of Fe, Ni, Al, Mn, Cu, Cr, and Co mixed with metalloids such as Si, Ge, As, B, C, P and N

Techniques for the preparation of metallic glasses:

There are several techniques available for the production of metallic glasses. The various techniques and the process involved in it are briefly listed in table.

Sl. No	Technique	Process
1.	Melt spinning system(Quenching technique)	Molten alloy is made to impinge on a fast rotating roller to form metallic glasses.
2.	Twin roller system	Molten alloy is passes through two rollers rotating in opposite direction to form metallic glasses.

3.	Melt extraction system	Fast moving roller sweeps off molten droplet into a strip to form metallic glasses.
4.	Sputtering	The sputtering gas is ionized and the atoms are made to fly towards the substrate to form metallic glasses.

Of these four techniques melt spinning (quenching) technique is commonly used and is discussed below.

5.2.1 PREPARATION OF METALLIC GLASSES

Principle

“Quenching” is a technique used to form metallic glasses, quenching means rapid cooling. Actually atoms of any materials move freely in a liquid state. Atoms can be arranged regularly when a liquid is cooled slowly. Instead, when a liquid is quenched, there will be an irregular pattern, which results in the formation of metallic glasses.

Technique

The process involved in the formation of metallic glasses is melt spinning technique. This technique is illustrated in figure 5.1.

Experimental setup

The setup consists of a refractory tube with fine nozzle at the bottom. The refractory tube is placed over the rotating roller made up of copper. An induction heater is wound over the refractory tube in order to heat the alloy inside the refractory tube as shown in figure 5.1.

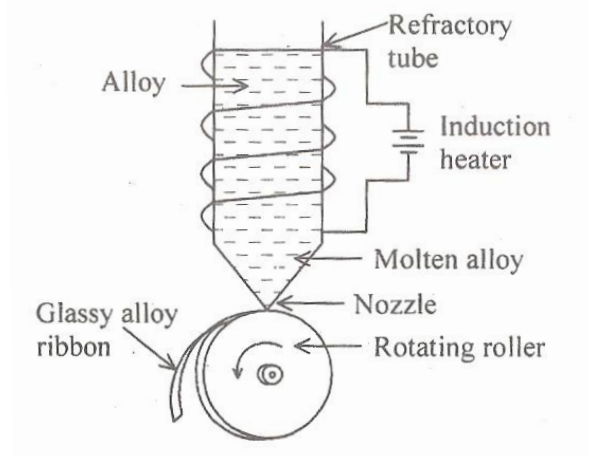


Fig 5.1 Production of Metallic glasses

Preparation

The alloy is put into the refractory tube and the induction heater is switched ON. This heats the alloy and hence the super-heated molten alloy is ejected through the nozzle of the refractory tube onto the rotating roller and is made to cool suddenly. The ejection rate may be increased by increasing the gas pressure inside the refractory tube. Thus due to rapid quenching a glassy alloy ribbon called metallic glass is formed over the rotating roller.

Types of metallic glasses

Metallic glasses are of two types viz.

(i) Metal-metalloid glass

Examples: metals : metalloids

 Fe, Co, Ni : Ge, Si, B, C

(ii) Metal-metal glasses:

Examples:	metals	:	metals
	Ni	:	Niobium
	Mg	:	Zn
	Cu	:	Zr

5.2.2 PROPERTIES OF METALLIC GLASSES

Since the atoms in the metallic glasses are disordered, they have some peculiar properties as follows.

(i) Structural properties:

- (a) Metallic glasses have tetrahedral closely packed (TCP) structure rather than hexagonal closely packed (HCP) structure.
- (b) They do not have any crystal defects such as grain boundaries, dislocations etc,

(ii) Mechanical (or) General Properties:

- (a) The metallic glasses are very strong in nature.
- (b) They have high corrosion resistance.
- (c) They possess malleability, ductility etc.,

(iii) Magnetic properties:

- (a) Metallic glasses can be easily magnetized and demagnetized.

- (b) They have very narrow hysteresis loop as shown in figure 5.2. In the hysteresis loop of the metal alloy in crystalline phase is also given for reference.
- (c) They exhibit very low hysteresis loss and hence transformer core loss is very less.

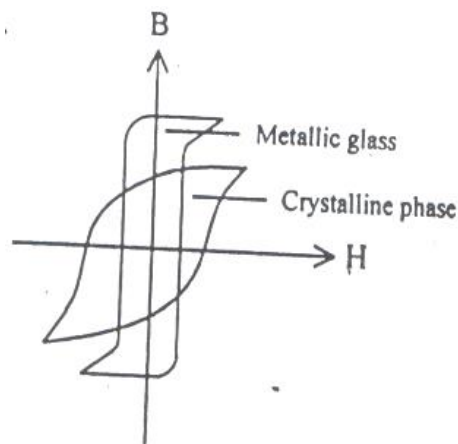


Fig 5.2 Hysteresis loop

(iii) Electrical properties:

- (a) Metallic glasses have high electrical resistance. ($100\mu\Omega\text{-cm}$).
- (b) The electrical resistance of metallic glasses will not vary with temperature.
- (c) They possess very low eddy current losses.

5.2.3 APPLICATIONS OF METALLIC GLASSES:

1. Since the metallic glasses possess low magnetic loss, high permeability, saturation magnetization and low coercivity, these materials are used in cores of high power transformer. Examples: $\text{Fe}_{75}\text{P}_{15}\text{C}_{10}$, $\text{Fe}_{24}\text{Zr}_{76}$, $\text{Ni}_{60}\text{Nb}_{40}$. etc.,

2. As the metallic glasses are malleable and ductile, it can be used in simple filament winding to reinforce pressure vessels.
3. Since the metallic glasses are very strong/hard they are used to make different kinds of springs.
4. As the met glasses are similar to the soft magnetic alloys, they are used in leads of tape record, core of transformer and magnetic shields.
5. Because of their high resistivity, they are used to make computer memories, magneto-resistance sensors etc.,
6. Since they have high corrosion resistance, they are used in reactor vessels, surgical clips, marine cables etc.,
7. Since some metallic glasses can behave as a superconductor, they are used in the production of high magnetic fields.
8. Since the met glasses are not affected by irradiation, they are used in nuclear reactors.

5.3 LIQUID CRYSTAL DISPLAY: [LCD] [Passive display device]

Liquid crystal display is not a semiconductor device as LED. LCD's display the light, it does not radiate light energy. Therefore LCD's require an external or internal source of light so that it can either transmit or reflect the incident light.

Definition

It is a digital display unit used in watches, calculators etc, in which the digits are displayed by the transmission or reflection of the incident light, with a very low power consumption (in the order of microwatts).

Theory

Liquid crystals are the intermediate phases between liquids and the crystals. It has fluidity and anisotropic properties. The liquid crystals are organic fluids. The individual molecule has a minute cigar shaped appearance. The molecule will have a preferred orientation or direction called director

Based on the orientation of the molecule there are three phases in liquid crystals

1. Smectic
2. Nematic and
3. Cholestric

Out of the three phases the last two phases are very important.

Nematic phase

In this phase the molecule are orderly arranged (i.e.) the directors are aligned parallel to each other as shown in the below figure 5.3.

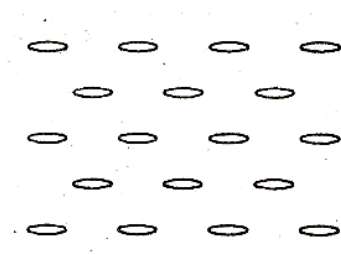


Fig 5.3 Nematic phase

The Nematic liquid crystal is made of MBBA molecule (i.e.) **4-methoxy-benzylidene, 4-butylaniline** in which two benzene rings are linked in a single group.

The chemical formula of the MBBA molecule is

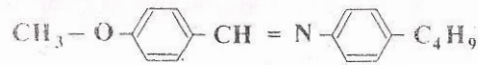


Fig 5.4 Chemical formula of MBBA molecule

Cholestric phase: In Cholestric the molecule are orderly arranged similar to Nematic phase but in planes. Here the only difference is the orientation (directors) will differ from one plane to another as shown figure 5.5.

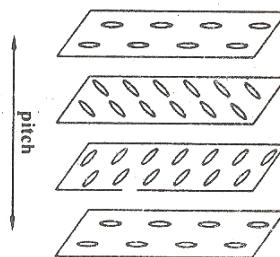


Fig 5.5 Cholestric phase

The distance between the similar planes (i.e.) the plane in which the molecule has same director is called pitch.

The Cholestric liquid crystal has a special property that if white light is allowed to pass through the crystal it appears coloured.

5.3.1 TYPE OF LCD'S

Nematic phase is widely used for displays. There are two types of LCD's in use,

1. Dynamic scattering displays
2. Twisted Nematic displays or field effect displays

5.3.2 DYNAMIC SCATTERING DISPLAY

In the case of dynamic scattering display in the absence of an applied voltage, the devices are transparent. But, when the voltage is applied, the liquid crystal becomes an efficient scatter of white light.

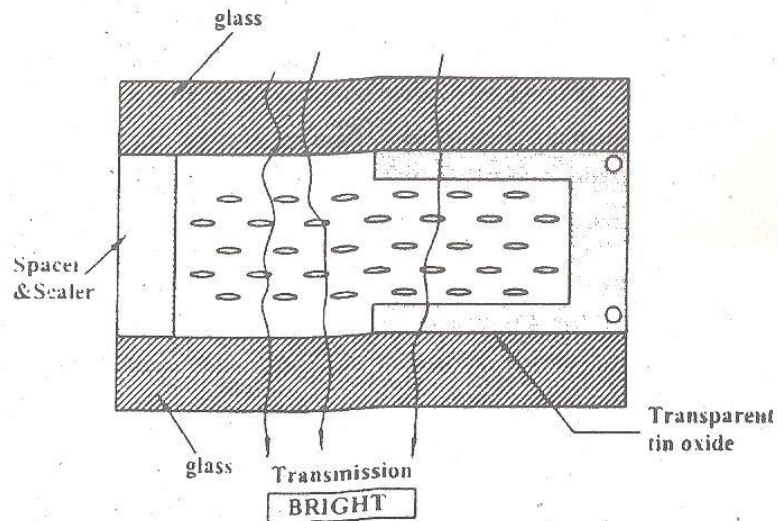


Fig 5.6 Without field

Construction

It consists of a pure Nematic liquid crystal in which impurities are added to increase its conductivity, due to ions in them. The liquid crystal is filled in between two glass plates which are coated with transparent tin oxide and can act as an electrode as shown in figure 5.6. The spacer and scalar help in filling of liquid and to change the liquid level.

Working

In the normal state without application of electric field, all the molecules in the Nematic liquid crystal are parallel to the glass plate. The material is transparent as shown in figure 5.6.

When the electric field is applied, the dipoles of the molecules are rotated into alignment in the direction, exactly perpendicular to field direction. Now the ions are pulled by the electrodes and hence the positive ions go towards negative potential and vice versa.

Therefore these ions (impurity) disrupt the molecules which are orderly arranged and create a small turbulence. Due to turbulence the negative charges are gathered as crests and positive charges are gathered as troughs, with respect to the field direction as shown in figure 5.7.

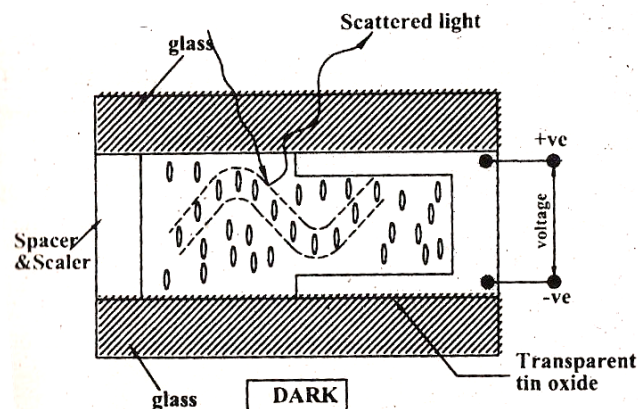


Fig 5.7 With field

When the applied voltage is at critical value, the turbulence becomes more, which results in the variation of refractive index of the medium of the crystal. So the light falling on them is scattered and hence liquid appears dark in the white background. When the field is switched off, the molecules are locally rearranged and materials become transparent and the display is erased.

Disadvantages

1. It has large operating voltage (15 volts).

2. It consumes large power (20 milliwatt).
3. Lifetime is very small.

5.3.3 THE TWISTED NEMATIC CRYSTAL DISPLAY

Construction

In this nematic phase, liquid crystal molecules are kept between two transparent glass plates. The top glass plate and the bottom glass plates are coated with a transparent electrode. The polarizer and analyzer are arranged in the crossed position, above and below the glass plates respectively as shown in figure 5.8.

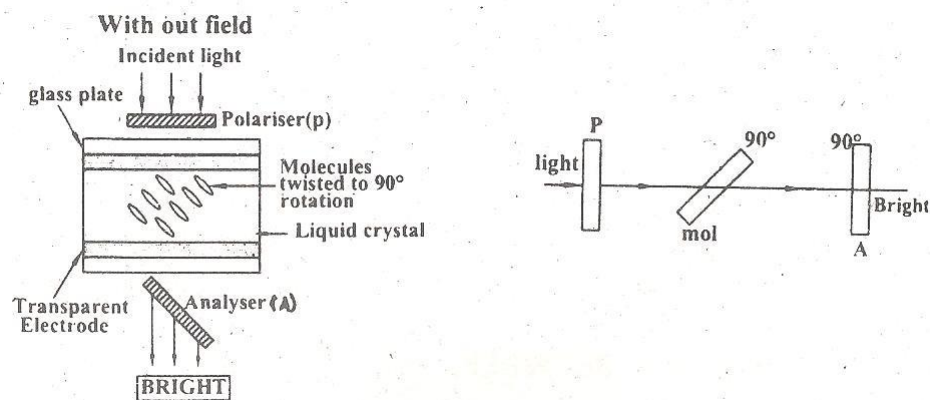


Fig 5.8 Without field

Working

The top glass plate is rotated through 90° and hence the molecules are also twisted through 90° (Fig 5.8).

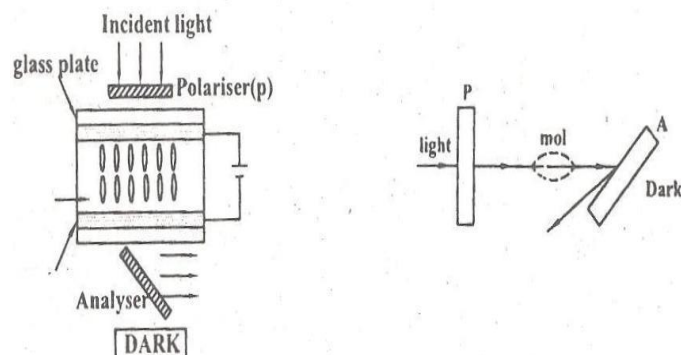


Fig 5.9 With field

Without field:

The light is passed through the polarizer, then through the molecules and finally through the analyzer.

Here the field of view appears bright, because the plane of polarization is rotated through 90° by twisted Nematic molecules (Fig 5.8).

With filed

Now when the field is applied, the twisted molecules come to normal position as shown in figure 5.9, and there is only 90° rotation of plane of polarization between polarizer and analyzer (Fig 5.9). Therefore, there is no transmission of light and the display appears black in white back ground.

When the electric field is switched off, once again the molecules return back to their twisted position and so a display is bright and erased.

5.4 Difference between LCD and LED

Sl.No	LCD	LED
1.	LCD is a liquid crystal display unit which modulates the light to have display.	LED is a light emitted diode which emits light during forward biasing.
2.	LCD is a passive display device	LED is an active display device.
	Advantages	Disadvantages
3.	Cost is very low	Cost is high
4.	Suitable for large area display	Not suitable for large area display
5.	Operating temperature is 10°C to 47°C	Operating temperature is 0°C to 70°C
6.	Response time is in micro seconds(10^{-6} sec)	Response time is in Nano seconds(10^{-9} sec)
7.	Intensity of light cannot be controlled	Intensity of light can be controlled.
8.	Colour displays will not be available at low cost	Different colour displays are available at low cost.

5.5 APPLICATIONS OF DISPLAY DEVICES:

1. They are used in wrist-watches, clock.
2. They are used in calculators, measuring instruments like digital multimeter.
3. Digital balance also has application of display devices.
4. They are used in measuring the temperature (digital thermometers).
5. They are used in digital gauss meters.
6. They are used in large display units.
7. They are used in power level indicator.

8. They are applied in optical switching devices.
9. LED's are used in dot matrix displays units.

SHAPE MEMORY ALLOYS

5.6 SHAPE MEMORY ALLOYS (SMA)

Definition

Shape memory alloys are the alloys which change its shape from its original shape to new shape and while heating/cooling it will return to its original shape.

5.6.1 TRANSFORMATION TEMPERATURE

Definition

The range of temperature at which the SMA switches from new shape to its original shape is called transformation temperature (or) memory transfer temperature.

- Below the transformation temperature the SMA can be bent into various shapes
- Above the transformation temperature the SMA returns to its original shapes
- The change in shape was mainly caused due to the change in crystal structure within the materials
- It occurs over a range of temperatures

5.6.2 PHASES (STRUCTURES) OF SMA

In general the SMA has two phases (crystal structures) viz.

1. Martensite (Fig 5.10)

2. Austenite (Fig 5.11)



Fig 5.10 Martensite

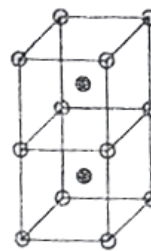


Fig 5.11 Austenite

Martensite

Martensite is an interstitial super solution of carbon in alpha-iron and it crystallizes into twinned structure as shown in figure 5.10. The SMA will have this structure generally at lower temperatures and it is soft in this phase.

Austenite

Austenite is the solid solution of carbon and other alloying elements in gamma-iron and it crystallizes into cubic structure in figure 5.11. The SMA will attain this structure at higher temperatures and it is hard in this phase.

5.6.3 PROCESSING OF SMA

Shape memory effect:

It is very clear that at lower temperature the SMA will be in Martensite structure and when it is heated then it will change its shape to Austenite structure

and while cooling it will again return to Martensite form. This effect is called shape memory effect.

Let us consider a shape memory alloy for which temperature is decreased. Due to decrease in temperature, phase transformation takes place from Austenite to twinned Martensite as shown in figure 5.12 (Process-1). i.e., a micro constituent transformation takes place from the platelet structure (Austenite) to needle like structure (Martensite).

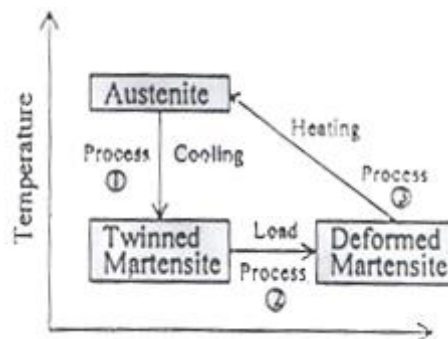


Fig 5.11 Phase transformation

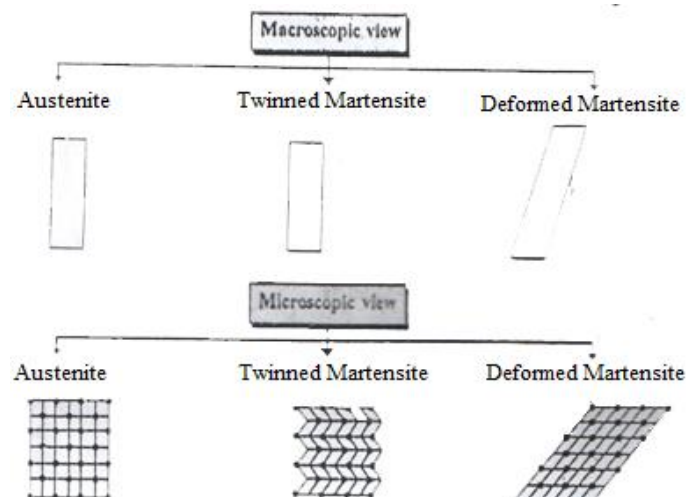


Fig 5.11 Deformation of SMA

During this state the twinned Martensite phase will have same size as that of Austenite phase as shown in below figure 5.13. Hence macroscopically if we see, no change in size (or) shape is visible between the Austenite phase and twinned Martensite phase of the SMA.

Both Austenite and twinned Martensite is stable in a particular range of temperature. Now when the twinned Martensite is applied a load, it goes to deformed Martensite phase as indicated in below figure 5.12 (Process-2). During the transformation from twinned Martensite to deformed Martensite the change in shape and size occur both microscopically and macroscopically as shown in below figure 5.13.

Now when the material is further heated it will go from deformed Martensite to Austenite form and the cycle continues as shown in below figure 5.12 (Process-3).

5.6.7 CHARACTERISTICS OF SMA

1. The transformation occurs not only at a single temperature rather they occur over a range of temperature.

2. Pseudo-elasticity

Pseudo-elasticity occurs in some type of SMA in which the change in its shape will occur even without change in its temperature. Say for example if we look into the stress-strain curve shown in figure 5.14, the SMA can change its shape from Austenite to Martensite during the transition temperature from T_1 to T_3 . At a constant temperature T_2 , when the stress is slightly increased a large strain is observed in the material and the phase transition takes place from Austenite to Martensite, which is indicated as line AB. Now when the stress is decreased the

material again goes to Austenite phase, as indicated by the line CD. Thus it can be noticed that the shape is recovered only due to stress and not due to temperature. This behaviour is called pseudo-elastic behaviour and the effect is called pseudo-elastic effect.

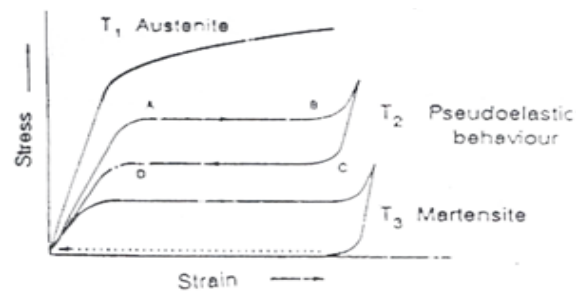


Fig 5.14 Stress-Strain curve

3. Super-elasticity

The shape memory alloys which have change in its shape at constant temperature are called super-elastic SMAs and that effect is known as super-elasticity.

Here, at a single temperature, when the load is applied the SMA will have a new shape and if the load is removed it will regain its original shape, similar to pressing a rubber (or) a spring.

4. Hysteresis

For an SMA, during cooling process, a Martensite starts (m_s) and ends (m_e) and during heating process, Austenite starts (A_s) and ends (A_e).

It is found that they do not overlap with each other and the transformation process exhibits the form of hysteresis curve as show in figure 5.15.

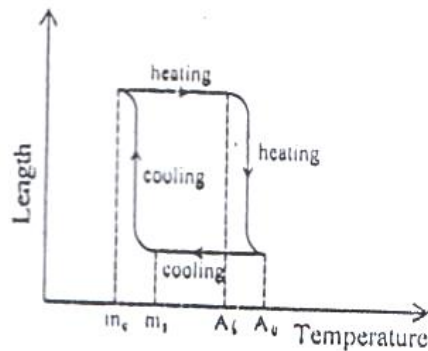


Fig 5.15 Hysteresis curve

5. Crystallographically the thermo-elastic Martensites are reversible.

5.6.5 TYPES OF SMA

There are two types of SMA viz.

- (i) One-way shape memory alloy (SMA)
- (ii) Two-way shape memory alloy (SMA)

(i) **One-way SMA:** From above figure 5.15 we can see that when the material is cooled below the Martensite ends (m_e) temperature, then there exist some minor change in energy inside the material, but the individual atomic units of Martensite will self-align in such a way that the material will remain in Martensite form alone.

Thus, though there is some change in its temperature, the SMA remains in the same phase, and this type of material is called one way shape memory alloy.

(ii) **Two-way SMA:** From above figure 5.15 we can note that if the temperature is increased then the Martensite becomes Austenite and if the temperature is decreased it reverses its states.

Thus the type of materials which produces spontaneous and reversible deformation just upon heating and cooling even without load are called two way shape memory alloys.

5.6.6 PROPERTIES OF NI-TI ALLOY

Nickel-Titanium [Ni-Ti] is type a shape memory alloy which has high shape memory strain.

It has the following properties, viz.,

- (i) Ni-Ti alloy has high shape memory strain [8.5%].
- (ii) The density of Ni-Ti is 6.45 gm/cm^3 .
- (iii) It is more flexible.
- (iv) It has high melting point [1300°C].
- (v) The transformation temperature varies between -200°C and 110°C .
- (vi) It has high thermal stability.
- (vii) It has high corrosion resistance.
- (viii) The thermal conductivity ranges from 8.5 [Martensite] to 18 [Austenite] $\text{W}^\circ\text{C/cm}$.
- (ix) It has very high yield strength of about 70 to 140 Mps in Martensite form, and about 200 to 700 Mps in Austenite form.

5.6.7 APPLICATION OF SMA

Shape memory alloys have vast application in our day-to-day life, as follows:

1. **Eye glass frames:** We know that the recently manufactured eye glass frames can be bent back and forth, and can retain its original shape within fraction of time. All these material are made up of Ni-Ti alloys, which can withstand to maximum deformation.
2. **Toys:** We might have seen toys such as butterflies, snakes etc., which are movable and flexible. These materials are made using SMAs.
3. **Helicopter blades:** The life time of helicopter blades depends on vibration and their return to its original shape. Hence shape memory alloys are used in helicopter blades.
4. **Blood-clot filter:** Here, the SMA is cooled and sent into vein. Due to body temperature its change its shape and acts as a blood clot filter, by which it controls the blood flow rate.
5. **Fire safety valves:** The SMA is mainly used to control and prevent the fire and toxic gases (or) liquids extent. For example, if an SMA is placed in a fire safety valve, when fire occurs, then due to change in temperature the SMA changes its shape and shuts off the fire. Similar principle has been used in the area of leakage in toxic gases (or) liquids.
6. **Coffee maker:** The Ni-Ti spring is used to release the hot milk and the ingredients at certain temperature and to close it after particular time, thereby we can get coffee automatically.

7. **Cryofit hydraulic couplings:** SMA is used for cryofit hydraulic couplings i.e., to join the ends of the tubes. Here, the SMA materials are pasted in between the two tubes to be joining at particular temperature when the temperature change the SMA expands and thus the two ends are joined.

8. **Circuit edge connector:** Using SMA the circuit can be connected and disconnected, depending on the variation in temperature. Hence SMA is used as a circuit edge connector.

9. **Cracks:** They are used in controlling and preventing cracks.

10. They are used in relays and activators.

11. They are used for steering the small tubes inserted into the human body.

12. They are used to correct the irregularities in teeth.

13. Ni-Ti SMA is also used in artificial hip-joint, bone-plates and pins for healing bones-fractures and also in connecting broken bones.

5.6.8 ADVANTAGES AND DISADVANTAGES OF SMA

Advantages

1. SMA is very compact in nature.
2. It is safe and smart.
3. They are flexible.
4. They are non-corrosive.

Disadvantages

1. Cost is high.
2. Efficiency is low.
3. Transformation occurs over a range of temperatures.
4. Structural arrangements may sometime get deformed.

5.7 NANOMATERIALS

Definition

Nano phase materials are the materials in which the atoms/grain size is in the order of 1 to 100 nanometers and these atoms will not have away from each other.

Egg:- ZnO, Cu-Fe alloys, Ni, Pd, Pt etc.

5.7.1 CLASSIFICATION OF NANOMATERIALS

The different forms of nano materials are classified on the basis of their dimensions along the three directions.

1. Zero dimensional
2. One-dimensional
3. Two dimensional
4. Three dimensional

Zero dimensional

Nano-materials are materials in which all the dimensions are measured within the nanoscale.

Eg:- Nanoparticles

1. They are amorphous or crystalline.
2. May be single crystal.
3. They exhibit various shape and forms.
4. They are metallic, ceramics or polymer.

One-dimensional

Nanomaterials differ from zero-dimensions if the materials have any of one dimension is outside the nanoscale i.e, greater than 100nm.then it is called one dimensional nanomaterial.

Eg:- Nanotubes, Nanorods and Nanowires.

Properties

1. They are amorphous or crystalline.
2. May be single crystal or polycrystalline.
3. They can be composed of single or multichemical elements.
4. They can be chemically pure or impure.
5. They are metallic, ceramics or polymer.

Two dimensional nanomaterials

Materials in which two of the dimensions are not confined to the nanoscale 2-D nanomaterials exhibit plate like shape.

Eg:- Nanomesh, Nanolayers, Nanocoating.

Properties:

1. 2D Nanomaterials can be amorphous or crystalline.
2. They are made up of various chemical compositions.
3. They may be used as single layer or multilayer.
4. They can be deposited on a substrate.
5. They are metallic ceramic or polymer.

Three dimensional nanomaterials

They are also known as bulk nanomaterials are materials which are not confined to nanoscale in any of the dimensions. These materials possess a nanocrystalline structure or exhibit the features at the nanoscale.

Eg:- Bundles of Nanowires and Nanotubes.

Properties

1. They may be amorphous or crystalline.
2. They can be chemically pure or impure.
3. They can composite materials.
4. They are metallic.

SYNTHESIS OF NANOMATERIALS**5.7.2 TOP-DOWN APPROACH**

In which bulk materials are broken into nanosize as shown in the figure 5.16.

BOTTOM UP APPROACH

In which nano materials are made by building atom by atom as show in figure 5.16.

Top down process: 1.Ball milling and 2.Laser ablation.

Bottom up process: 1.CVD, 2.PVD and 3.Sol-Gel method.

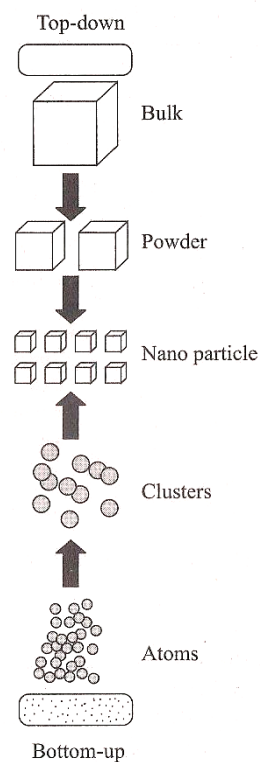


Fig 5.16 Top-down method

5.7.3 PVD OR PHYSICAL VAPOUR DEPOSITION OR EVAPORATION:

PVD is a process of transferring growth species from a solid material and deposit them on a substrate to form a film. In this process the growth species are removed from the target material by resistive heating or electron beam heating. The system consists of a evaporation source that vaporises the desired material.

The substrate placed at an appropriate distance facing the evaporation source. Both source and the substrate are located in a vacuum chamber. The substrate can be heated or cooled during deposition. The deposition of this film is carried out in a chamber with a low pressure $< 10^{-7}$ torr.

The presence of vacuum in the chamber prevent collision of vapourised ions with each other the vacuum also prevents contaminants the required vacuum in the system is obtained using vacuum pumps such a diffusion and turbo molecular pumps with the support of rotary pump to prepare nanomaterials the desired material taken in the tungsten or molybdenum boat is evaporated using resistive or electron beam heating inside the vacuum chamber.

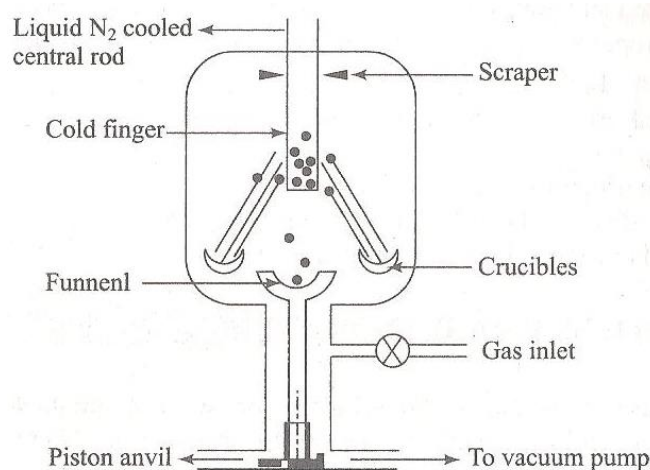


Fig 5.17 Physical vapour phase-evaporation.

Now the evacuated chamber is filled with inert gas. The metal vapour migrates from the hot source into the cooler inert gas and the evaporated atoms collide with the gas atoms within the chamber and lose its kinetic energy.

Thus the metal vapour cools through collisions with inert gas species and becomes supersaturated the super saturated gas species then nucleates

homogenously with particle size ranging from 1 to 100nm the particle size can be controlled by the inert gas pressure.

5.7.4 LASER ABLATION TECHNIQUE

The laser ablation technique as show in figure 5.18 consists of a high intense laser beam, a cylindrical furnace, an air or water cooled metallic trap with filler, a vacuum pump and an inlet for the gas. A CO₂ laser with an output wavelength of 10.6 micrometer and power output of 1 KW can be focused into small spot in the order of mm using a lens produces a very high energy, and it is made to incident on a target material. If the temperature produced by the laser beam on the target material is greater than its sublimation point, the explosion of the target material may occur and hence the target material gets efface from the surface. The nanomaterials are collected on the air or water cooled traps. A continuous flow of nitrogen gas with a pressure of nearly 1 Torr is maintained during the synthesis of the nanomaterials. A tube furnace is an additional energy source used to maintain the temperature of the target. This method is known as laser ablation.

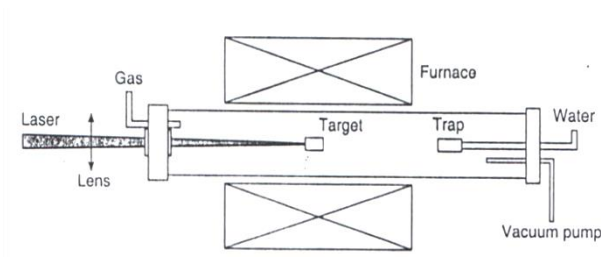


Fig 5.18 Laser ablation

If the temperature produced by the laser beam on the target material is less than its sublimation point, the laser radiation heats the target material without the obvious ejection of the target material. This method is said to be

laser heating. Both laser ablation and laser heating can be used for producing nanomaterials.

Advantage

1. The nanomaterials with perfect structures can be produced.
2. Both single wall and multiwall nanotubes can be produced.
3. The process is easily controllable by controlling the temperature, pressure and laser output power.

Drawbacks

1. High temperature should be maintained.
2. Laser source with high output energy should be needed.
3. Only a small quantity of nanomaterials can be produced.

5.7.5 BALL MILLING AND ANNEALING

This method is a two-step process. The first step is ball milling at room temperature and another step is annealing at relatively low temperatures. These two processes correspond to separate nucleation and growth process.

A ball mill consists of a stainless steel container and several iron balls or silicon carbide balls, the material powder is taken in the ball milling chamber in addition to the several hundreds of silicon carbide balls. An external magnet introduces the pulling forces to the materials and hence it increases the milling energy. When the milling chamber rotates the balls, they drop to bottom due to the gravitational force and due to the magnetic force they reach the top of the chamber and hence the balls move continuously as shown in figure 5.19. The

silicon carbide balls impart very high amount of energy to the material powders and hence the powders get crushed. The milling is to be done nearly 100 h to 150 h to get uniform fine powders. After preparing fine powders, annealing is done at a suitable temperature for a certain period of time so as to get nanotubes.

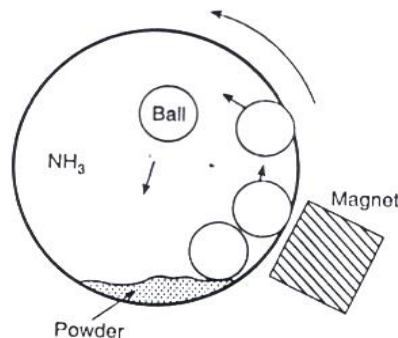


Fig 5.19 High energy ball mill (HEBM) setup

The milling process produces a number of operations such as fracturing, grinding, high speed plastic deformation, cold welding thermal shock, intimate mixing etc of the materials. In ball milling, the structural and the chemical changes are produced by the mechanical energy rather than the thermal energy at low temperature.

Advantages

1. It is a well-defined, inexpensive process.
2. Nanopowders with particle size 2 to 20nm in large quantities can be produced.

Disadvantages

1. The shape of the nanoparticles is irregular.
2. This method introduces crystal defects.

3. Introduction of impurities from balls and milling additives.

5.7.5 SOL-GEL PROCESS

The sol-gel process is a wet-chemical method used to produce high purity oxide nanoparticles such as TiO_2 , ZnO_2 , etc. Suitable precursors in the form of nitrates/acetates/carbonates are taken and then dissolved in deionized water. The solutions are kept at suitable temperatures. The starting material is used to produce a colloidal suspension known as sol. Then a gelling agent like PVA is added. This will produce a gel. A thin film coating is made on a substrate such as glass or Ti or Ni sheets depending upon the requirements using sol-gel dip-coating or spin-coating. To make a uniform coating the pH, temperature and viscosity should be controlled. Then the film is annealed at suitable temperature and then studied.

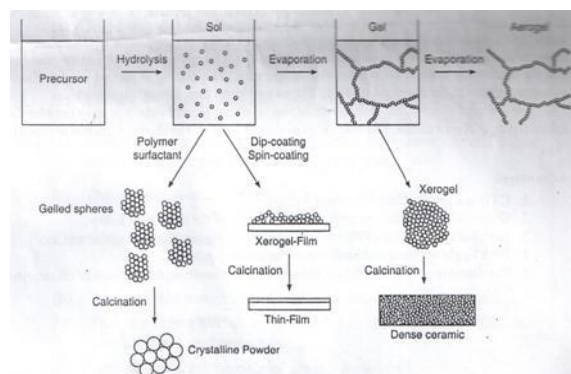


Fig 5.20 Sol-Gel Process

Instead of forming a thin film coating, crystalline powders are also produced by suitably calcining the gel. The sequence of steps involved in a sol-gel process is displayed in figure 5.20.

Advantages

1. It is a low-cost method used to produce large quantities of nanopowders.
2. The sol-gel process can be done at low temperature
3. This method produces homogeneous substances

5.7.6 CHEMICAL VAPOUR DEPOSITION (CVD)

Chemical vapour deposition is a chemical method used to produce nanomaterials. The CVD apparatus consists of a quartz tube container, which is heated by a tubular furnace as show in figure 5.21.

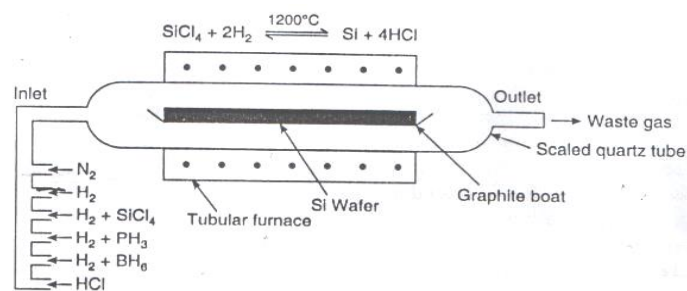
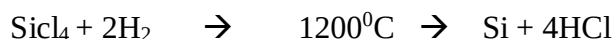


Fig 5.21 CVD process

A silicon substrate is kept inside the quartz container in a tungsten/graphite boat. The chamber is evacuated and an inert gas is passed into the chamber. The reactants are admitted into the chamber and the chamber is maintained at a suitable temperature. The coating is formed on the silicon substrate because of the chemical reaction. The unused gases flow through the outlet.

The chemical reaction, the hydrogen reduction of silicon tetrachloride, is used to produce the epitaxial growth of pure silicon. This reaction takes place at 1200°C . The chemical reaction involved is

**Advantages**

1. CVD is a low-cost and high-yield method.
2. Compared to the other methods, the temperature of the deposition is low.
3. High purity (more than 99% pure) nanomaterials are produced using this method.
4. Both single wall and multiwall nanotubes can be produced.
5. The diameter of the nanotube can be controlled by controlling the thickness of the catalytic film.

5.7.7 PROPERTIES OF NANO-PARTICLES

As the size of the nanoparticle is very less, these materials exhibit peculiar physical, electronic, magnetic, mechanical and chemical properties. Some of them are discussed below.

1. Physical properties

- (a) Since the size of the particle is very less, the particles are very close to each other and hence the inter particle spacing is very less in nano-materials.
- (b) Because of its very less size, these nano-materials cannot be further divided into smaller particles and it does not have any dislocation in it. Thus we can say that they have high strength and super hardness.
- (c) The melting point of nano-materials will be very less.

2. Electronic properties

- (a) The energy bands in these materials will be very narrow.
- (b) The ionization potential is found to be higher for nano-materials.
- (c) When the nano-materials are prepared from bulk materials, they have more localized molecular bonds.
- (d) Nano-materials are capable of storing hydrogen atoms.

3. Magnetic properties

- (a) In nano-materials a large number of atoms will be present at the surface. These atoms will have less co-ordination number and hence possess local magnetic moment within them.
- (b) Due to large magnetic moment these nano-materials exhibits spontaneous magnetisation at smaller sizes
- (c) Ferro-magnetic and anti-ferro magnetic multi-layer nano-materials has GMR (Giant Magneto Resistance) effect.
- (d) The nano-materials shows variation in their magnetic property, when they change from bulk state to cluster (nano particle) state. Some of the examples are listed in table

Sl.No	Material	Bulk state	Nanophase state
1.	Iron, Ni, Cobalt	Ferro magnetic	Super para-magnetic
2	Sodium, Potassium	Para magnetic	Ferro magnetic
3.	Chromium	Anti-Ferro magnetic	Frustrated paramagnetic

4. Mechanical properties:

- (a) The hardness of nano-phase materials varies from material to material. This may be due to the phase transformation, stress relief, density and grain boundaries.
- (b) They exhibit super plastic behavior.

5.7.8 APPLICATIONS OF NANO-PARTICLES

Though nano-particles are very small, they are the important materials to build the future world. They have applications almost in all engineering fields as follows:

1. Mechanical engineering

- (a) Since they are stronger, lighter etc., they are used to make hard metals.
- (b) Smart magnetic fluids are used in vacuum seals, magnetic separators etc.
- (c) They are also used in Giant Magneto Resistance (GMR) spin valves.
- (d) Nano-MEMS (Micro-Electro Mechanical Systems) are used in ICs, optical switches, pressure sensors, mass sensors etc.

2. Electrical, electronics and communication engineering

- (a) Orderly assembled nano-materials are used as quantum electronic devices and photonic crystals.
- (b) Some of the nano-materials are used as sensing elements. Especially the molecular nano-materials are used to design the robots, assemblers etc.

- (c) They are used in energy storage devices such as hydrogen storage devices, magnetic refrigeration and in ionic batteries.
- (d) Dispersed nano-materials are used in magnetic recording devices, rocket propellant, solar cells, fuel cells, etc.
- (e) Recently nano-robots were designed, which are used to remove the damaged cancer cells and also to modify the neuron network in human body.

3. Computer science engineering and IT

- (a) Nano-materials are used to make CD's and semiconductor laser.
- (b) These materials are used to store the informations in smaller chips.
- (c) They are used in mobiles, lap-tops etc,
- (d) Further they are used in chemical/optical computers.
- (e) Nano-dimensional photonic crystals and quantum electronic devices plays a vital role in the recently developed computers.

4. Bio-medical and chemical engineering

- (a) Consolidated state nano-particles are used as catalyst, electrodes in solar and fuel cells.
- (b) Bio-sensitive nano-particles are used in the production of DNA-chips, bio-sensors etc.
- (c) Nano-structured ceramics materials are used in synthesis bones.
- (d) Few nano-materials are also used in absorbents, self cleaning glass, fuel additives, drugs, ferro fluids paints etc.
- (e) Nano-materials colloids are used as film precursors.

5.8 CARBON NANOTUBES (CNT)

Definition

Carbon nanotubes (CNT) are molecular scale tubes of graphitic carbon with outstanding properties they are among the stiffest and strongest fibres researched till date, with remarkable electronic properties and applications.

Types

Carbon nano-tubes are of two types:

1. single walled nano-tubes (SWNT`s)
2. Multi-walled nano-tubes (MWNT`s)

5.8.1 STRUCTURE OF THE CARBON NANOTUBE

It is possible to explain the different structures by considering the way in which the graphite sheet might be rolled into tubes.

Let us consider the structure of graphite sheet as shown in figure 5.22.

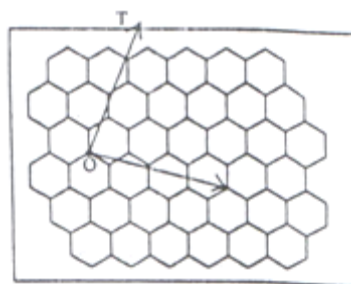


Fig 5.22 Graphite sheet

When the graphite sheet is rolled up about an axis T, the CNT is formed. The circumferential vector is at right angles to T. By rolling the graphite sheet

about the T vector at different orientations, three different CNT structures can be obtained. When the vector is parallel to the C-C bond of the carbon hexagons a structure known as armchair structure is obtained. By rolling the graphite sheet at different orientations about a T vector, Zigzag and chiral structures are obtained. In both cases the T vector is not parallel to C-C bonds.

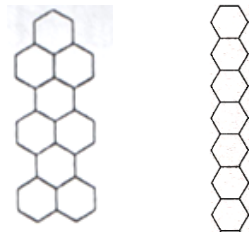


Fig 5.22 CNT Arm chair & zigzag structures

5.8.2 FABRICATION OF CARBON NANOTUBES

Carbon Nano-tubes can be fabricated by anyone of the following

- Pulsed laser deposition (or) Laser synthesis.
- Carbon arc method.
- Chemical vapour deposition.

5.9 PULSED LASER DEPOSITION

Pulsed laser deposition or Laser synthesis is the laser technique adopted for the preparation of carbon nano tubes.

Principle

The technique of laser heat treatment is used in the preparation of carbon nano-tubes. In general ruby laser, Nd-YAG laser are used for this purpose.

Instrumentation

The instrumentation for the fabrication of carbon nanotubes consists of a quartz tube containing a graphite target kept in argon gas region as shown in figure 5.24.

The tube is surrounded by an electric furnace in order to heat the target. A colder copper collector rod used to collect the nano carbons emitted by the graphite.

Along with this it also consist of a pulsed laser source to produce laser beam, shutter to control the intensity of the laser beam and an assembly of lenses to effectively focus the laser on to the graphite.

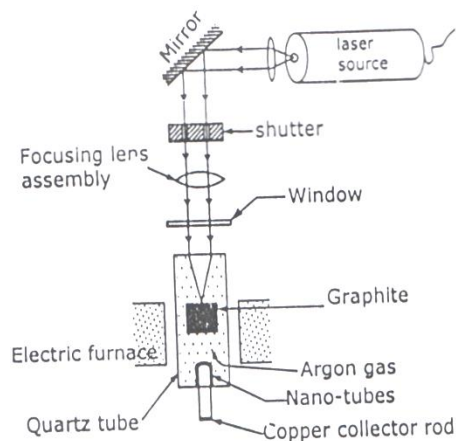


Fig 5.24 Pulsed laser deposition

Synthesis

Initially the graphite is heated up to 1200°C with the help of the electric furnace. An intense laser beam can be used to evaporate carbon from the graphite and thus now the laser source is switched ON.

The light reflected by the plane mirror is made to pass through the shutter. The intensity of the laser beam is controlled by the shutter and the controlled laser beam is allowed to fall on the focusing lens assembly. This lens assembly focuses the light effectively on to the window and is made to incident on the graphite.

Due to laser heating the graphite gets heated and evaporates carbon atoms. The argon gas present inside the quartz tube is used to sweep the carbon atoms towards the colder copper collector rod. Due to the movement of carbon atoms from a higher temperature region to lower temperature region it gets condensed and hence carbon nanotubes over the collector rod.

The Cobalt and Nickel present in the graphite act as catalytic nucleation sites for the formation of carbon nano tubes.

5.10 CAREBON-ARC METHOD

Carbon arc method is similar to that of electro-deposition technique and is used to produce both the single walled nanotube [SWNTs] and multiwalled nano tubes [MWNTs]

Instrumentation

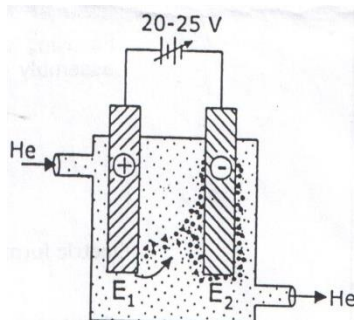


Fig 5.25 Carbon arc cell

It consists of positive carbon electrode E_1 and negative carbon electrode E_2 as shown in figure 5.25. Liquid helium of 500 Torr pressure is made to flow through the electrodes to deposit the carbon from E_1 to E_2 .

Fabrication

The battery is switched on and a potential of about 20 to 25 volt is applied across the carbon electrodes. This potential leads to the emission of carbon atoms from the positive electrode. Now with the help of high pressure Helium, the carbon atoms are deposited over the negative electrode.

Now when the potential is continuously applied, the length of the positive electrode start decreasing and the carbon deposit over the negative electrode starts increasing which results in the formation of carbon nano tubes layer by layer, so called multi walled carbon nano tubes MWNTs.

NOTE: The single walled carbon nano tube [SWNT] can also be produced by the above technique just by using cobalt, nickel or iron as a catalyst in the central region of the positive electrode.

5.11 CHEMICAL VAPOUR DEPOSITION TECHNIQUE:

In this method a hydrocarbon gas such as methane is heated up to 1100°C and is decomposed. As the gas decomposes, it produces carbon atoms. These carbon atoms are made to deposit over a cooler substrate, which contains iron as a catalyst to form carbon nanotubes.

NOTE: Using this technique we can fabricate carbon nanotubes continuously.

5.12 PROPERTIES OF CARBON NANO-TUBES

- The strength of the sp^2 carbon-carbon bonds gives high hardness for carbon nanotubes.
- The Young's modulus of the nanotubes is approximately 5 times greater than the steel.
- The tensile strength is around 50 times higher than steel.
- The electronic properties of carbon nanotubes are also extraordinary. Especially the Carbon nanotubes can be obtained in the form of metallic or semiconducting, based on its structural content.
- Nanotubes have higher conductivities than that of copper.

5.13 APPLICATIONS OF CARBON NANOTUBES

- The carbon nanotubes are very light in weight but they are very strong hence they are used in aerospace.
- They are used in constructing nano scale electronic devices
- Carbon nanotubes [CNTs] are used in battery electrodes, fuel cells, reinforcing fibers etc.
- CNTs are used in the development of flat panel displays for computer monitors and televisions

- Plastic Composite CNTs are used as a light weight shielding materials for protecting electromagnetic radiation.
- Light weight CNTs are also used in military and communication systems for protecting computers and electronic devices.
- Semiconducting CNTs are used as switching devices.
- Semiconducting CNTs are also used as chemical sensors to detect various gases.
- Nano-tubes can also serve as catalysts for some chemical reactions.
- The unique properties of carbon nanotubes will undoubtedly lead to many more applications in future to produce nano computers plastic composites etc.

Pondicherry University 2 marks question and answer**1. What are the basic types of ordering in liquid crystals? (April / May 2011, April1, Jan13, May14, Jan15,)**

Based on the orientation of the molecule there are three phases in liquid crystals

1. Smectic
2. Nematic and
3. Cholestric

2. What is meant by nanophase materials? (April/May 2011)

The Nanostructured materials with ultra-fine grain size in order of 1 to 100 nm are called as nanophase materials.

3. What are metallic glasses? (April10)

Metallic glasses are metal alloys which have no long range atomic order, high strength, good magnetic properties, better corrosion resistance and ease of fabrication. To achieve this structure a suitable alloy must be quenched so rapidly that the random liquid like structure is retained in the solid

4. How do nano materials differ from macro materials? (Jan11)

The Nanostructured materials with ultra-fine grain size in order of 1 to 100 nm are called as nanophase materials.

5. State any two applications of CNT. (Jan11)

- Hydrogen storage

- Rechargeable lithium batteries

6. State the properties of metallic glasses. (Jan14)

- Metallic glasses have tetrahedral closely packed (TCP) structure rather than hexagonal closely packed (HCP) structure.
- The Metallic glasses are very strong in nature.
- They possess malleability, ductility etc.
- They exhibit very low hysteresis loss and hence transformer core loss is very less.

7. Mention few techniques for synthesis of nanophase materials. (Jan14)

Top-Down approach:

1. Mechanical Alloying
2. Lithography
3. Erosion etc.

Bottom-Up approach:

1. Chemical precipitation
2. Gas Phase agglomeration
3. Sol-gel method etc.

8. What are the different forms of nanomaterials? (Nov11)

Sl.No	Types	Crystalline/grain size/shape.
1.	Zero dimension	Clusters or powdered(quantum dots)
2.	One dimension	Multilayer
3.	Two dimensions	Ultrafine grained overed layers or buried layers
4.	Three dimension	Equiaxed Nano-meter sized grains.

9. Mention any two physical properties of carbon nano tubes (CNT). (Nov11.May14)

1. The young's modulus of the carbon Nano tube is in the range from 1.28 to 1.8 Tpa. ($1\text{Tpa} \sim 10^7\text{atm}$). While for steel is 0.21 Tpa. It means that the CNT is 10 times stronger than that of steel.

2. A small amount of electric field is applied parallel to the axis of a Nano tube. As a result it emits electrons at a very high rate from the ends of the tube. This emission is known as field emission. The electron emissions of carbon Nanotube are used in electronic industries like flat panel display.

10. What are the applications of metallic glasses? (Jan12)

1. Since the metallic glasses possess low magnetic loss, high permeability, saturation magnetization and low coercivity, these materials are used in cores of high power transformers.
2. Since the metallic glasses are very strong and hard they are used to make different kinds of springs.
3. Because of their high resistivity, they are used to make computer memories, magneto-resistance sensors etc.
4. Since the metallic glasses are not affected by irradiation, they are used in nuclear reactors.

11. Mention the physical properties of carbon nanotubes? (Jan12)

- i. Carbon nanotubes are long , thin cylinder discovered by S.Iijima in1991
- ii. They a broad range of electronic , thermal and structural properties that change depending on the different kinds of nanotubes.

12. Mention the fabrication techniques of CNT? (Jan13)

- (i) PVD (ii) Chemical vapour deposition (iii) Electro-deposition

13. What is Shape memory effect? (May14)

It is very clear that at lower temperature the SMA will be in Martensite structure and when it is heated then it will change its shape to Austenite structure and while cooling it will again return to Martensite form. This effect is called shape memory effect.

14. Mention the different structures of CNT? (May14)

Carbon nano tubes are having three different structures depending upon the orientation of C-C bonds. They are

1. Armchair structure
2. ZigZag structure
3. Chiral Structure

15. Define CNT. What are the types of CNT? (May15)

Carbon nano tubes are sheet tubes formed by rolling the graphite (or graphene) sheet into tubes with the bonds at the end of the sheet. These bonds are used to close the tube. They posses exotic properties like very low specific resistivities in metallic carbon nano tubes and high mobilities for semi conducting nanotubes.

The two types of Carbon nano tubes are

1. Multi walled carbon nanotubes (MWCNT)
2. Single walled carbon nanotubes (SWCNT)

Additional 2 marks question and answer

Part - A

1. What are zero dimensional nanomaterials? Give examples.

Nanomaterials differ from zero-dimensions if the materials have any of one dimensions is outside the nanoscale i.e, greater than 100nm, then it is called one dimensional nanomaterial.

Eg:-nanotubes, nanorods and nanowires.

2. What are two dimensional nanomaterials? Give examples.

Materials in which two of the dimensions are not confined to the nanoscale 2-D nanomaterials exhibit plate like shape.

Eg:-nanomesh, nanolayers, nanocoating

3. Distinguish between active and passive display units?

Active display device: These devices display or emit the light radiation in their own due to application of field.

Example: LED – Light Emitting Diode

Passive display device: These devices modulate the incident radiation and then collect or transmit the light.

Example: LCD- Liquid Crystal Display

4. List out the differences between LED and LCD.

Sl.No	LED	LCD
1	Cost is high.	Cost is very low.
2	Not suitable for large area display.	Suitable for large area display.
3.	High consumption power (Milliwatts).	Low power consumption (microwatts)
4.	Response time is in nano seconds (10 ⁻⁹ seconds).	Response time is in microseconds (10 ⁻⁶ seconds).
5	Intensity of light can be controlled	Intensity of light cannot be controlled
6	Different colour displays are available at low cost.	Colour displays will not be available at low cost.
7	Operating temperature is 0° to 70°C	Operating temperature is 10°C to 47°C

5. What is meant by LCD? Give its principles and types.

LCD is the abbreviation of Liquid Crystal Display. It is a digital display unit in which the digits are displayed by the principle of transmission or reflection of the incident beam.

Types: (i) Dynamic scattering display.

(ii) Twisted nematic display.

6. What is a liquid crystal? Give its various phases.

Liquid crystals are organic fluids in cigar shape and have the phases between liquid and a crystal. The various phases of liquid crystals are

(i) Smectic phase (ii) Nematic phase

7. What is meant by MBBA molecule? Give its structure.

4-Methoxy -Benzylidene, 4- Butylaniline is called as MBBA molecules. It is formed by two benzene rings linked in a single group.

8. Define Director and Pitch.

Director: In liquid crystals the molecules will have a preferred orientation or direction is called director

Pitch: The distance between two similar planes in which the molecule has same director is called pitch.

9. Out of LED and LCD which is more advantages? Why?

Out of LED and LCD, LCD is more advantages because of the following reasons

- (i) It has low cost
- (ii) It has low power consumption
- (iii) It occupies smaller area
- (iv) It has many applications in all digital equipments.

10. Give any four applications of LCD, (Or) what are the applications of display devices?

- (i) They are used in wrist-watches, clocks.
- (ii) They are used in calculators, measuring instruments like digital multimeter.
- (iii) Digital balance also has application of display devices.
- (iv) They are used in measuring the temperature (Digital Thermometers).
- (v) They are used in Digital Gauss Meters.

11. What is meant by glass transition temperature?

The temperature at which the metals (alloys) in the molten form transform into glasses i.e. liquids to solids is known as glass transition temperature (T_g).

12. What do you understand by the term Quenching?

‘Quenching’ is a technique used to form metallic glasses, quenching means rapid cooling. Actually atoms of any materials move freely in a liquid state. Atoms can be arranged regularly when a liquid is cooled slowly. Instead, when a liquid is quenched, there will be an irregular pattern, which results in the formation of metallic glasses.

13. What are the types of metglasses?

Metallic glasses are of two types

(i) Metal-metalloid glasses

Examples: Metals- Fe, Co, Ni

Metalloids – Ge, Si, B, C

(ii) Metal-Metal glasses

Examples: Metals-Ni, Mg, Cu

Metals – Niobium, Zn, Zr

14. Mention any four properties of metglasses?

(i) Metallic glasses have tetrahedral closely packed (TCP) structure rather than hexagonal closely packed (HCP) structure.

(ii) The Metallic glasses are very strong in nature.

(iii) They possess malleability, ductility, etc.

(iv) They exhibit very low hysteresis loss and hence transformer core loss is very less.

15. Write any four applications of metglasses.

1. Since the metallic glasses possess low magnetic loss, high permeability, saturation magnetization and low coercivity, these materials are used in cores of high power transformers.
2. Since the metallic glasses are very strong and hard they are used to make different kinds of springs.
3. Because of their high resistivity, they are used to make computer memories, magneto-resistance sensors etc.
4. Since the metglasses are not affected by irradiation, they are used in nuclear reactors.
5. Dielectric strength: When external field applied to a dielectric material is greater than the critical field, the dielectric loses its insulating property and becomes conducting. Therefore a large current flows through the material. This phenomenon is called dielectric breakdown.

16. What are nano-phase materials? Give examples.

Nano-phase materials are the materials in which the grain size is in the order of 1 to 100 nano-meters and these atoms will not move away from each other. They exhibit greatly altered Physical, Chemical and Mechanical properties compared to their normal, large grained counterparts with the same chemical composition.

Examples: ZnO, Cu-Fe alloys, Ni, Pd, Pt etc.

17. Give any four techniques adopted to synthesize the nano-phase materials.

- (i) Mechanical crushing or ball milling
- (ii) Chemical vapour deposition
- (iii) Electro-deposition
- (iv) Laser synthesis

18. Classify Nanomaterials based on their grain size.

Sl.No	Types	Crystallite / grainsize / shape
1.	Zero dimension	Clusters or powders (quantum dots)
2	One dimension	Multilayer
3	Two dimension	Ultra fine grained over or buried layers
4	Three dimension	Equiaxed nanometer sized grains

19. What are the two ways adopted to synthesis nano particles?

The two ways of synthesizing nano particles are

- (i) Top-Down approach
- (ii) Bottom-Up approach

20. What are different types techniques used in Top-Down and Bottom-Up approach for synthesizing nano materials?

Top-Down approach:

- 1. Mechanical Alloying
- 2. Lithography
- 3. Erosion etc.,

Bottom-Up approach:

- 1. Chemical precipitation.
- 2. Gas Phase agglomeration.
- 3. Self assembly etc.,

21. What are the important properties of nano materials?

- (vi) The size of the nano-material will be in the order of 1-100nm and hence the inter particle spacing is found to be very less and they do not have any dislocation in it.
- (vii) Nano-materials have high strength and super hardness.
- (iii) The energy bands in these materials will be very narrow and the ionization potential is found to be higher.
- (viii) Nano-materials possess local magnetic moment within themselves, and these materials exhibit spontaneous magnetization at smaller sizes.

22. List out any four recent applications of nano- materials

- (i) Nano-MEMS (Micro-Electro Mechanical Systems) are used in ICs, Optical Switches, Pressure sensors, mass sensors etc.
- (ii) Recently nano-robots were designed, which are used to remove the damaged cancer cells and also to modify the neuron network in human body.
- (iii) Nano-dimensional photonic crystals and quantum electronic devices play a vital role in the recently developed computers.
- (iv) Bio-sensitive nano-particles are used in the production of DNA-chips, bio-sensors etc.

24. What is the difference between Nano materials and Macro materials?

All materials are composed of grains, which in turn comprise many atoms. Depending on the size these grains may be visible or invisible to the naked eye. Conventional materials or Macro materials have grain of size varying from hundreds of microns to centimeters. Nanomaterials have grain size in the order 1-100nm. Such grains contain maximum of few hundreds of atoms.

24. What are Carbon Nano tubes or CNT?

Carbon nano tubes are sheet tubes formed by rolling the graphite (or graphene) sheet into tubes with the bonds at the end of the sheet. These bonds are used to close the tube. They possess exotic properties like very low specific resistivities in metallic carbon nano tubes and high mobilities for semi conducting nanotubes.

25. List out the types of CNT

The two types of Carbon nano tubes are

1. Multi walled carbon nanotubes (MWCNT)
2. Single walled carbon nanotubes (SWCNT)

26. Mention any some important applications of Carbon nano tubes.

- (ix) The nanotubes are used for electron emission effect in vacuum lamps. The resultant lights are much brighter than the conventional light.
- (x) Nano metal wires with small diameters are used to do the interconnections between the computers and switches.
- (xi) Carbon nano tubes are used as a computer switching device.
- (xii) Carbon nanotubes have been used as a storage device in battery, fuel cells, etc.,
- (xiii) Chemical sensors are made using CNT to detect various gases like NO_2 .
- (xiv) CNT is used as a catalyst for enhancing the chemical reactions.

Part –B

Pondicherry University 11 marks question

1. How do Nanomaterials differ from bulk materials? Explain the preparation of nanomaterials by any two methods and give their properties. (Jan13, May14, May14r)
2. What are metallic glasses? How are they prepared? Explain their properties and applications. (May11, April10, Jan11, Jan12, Jan13, May14, May15, May14r)
3. Discuss the physical and applications of carbon nano tubes. (April11, April10, Jan11, Jan14, May14, May14,)
5. What is liquid crystal? Explain the types of liquid crystal display and its applications. (Jan11, Jan14, Jan 11, Jan12).
4. Explain briefly Ball milling process. (Jan15)
5. Explain briefly about Top down method for synthesis of nanomaterials.
6. What are the basic chemical reactions involved in CVD process? Give example in each case. (Jan15)